

Define Problem Statements Template

Date	15 March 2026
Team ID	Batch 2023-27 ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	3 Marks

Step 1: Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

Problem Statement Definition

A clear problem statement focuses the team on the core challenge and guides the design of the solution.

Attribute	Details
Problem Statement	Floods are among the most devastating natural disasters, claiming thousands of lives and displacing millions each year. Conventional forecasting methods lack ML-powered real-time predictive intelligence, leaving authorities with insufficient lead time to respond.
Who is affected?	Disaster management officers, meteorologists, local authorities, and flood-affected communities.

Attribute	Details
What is the gap?	No accessible, ML-powered web application that accepts meteorological inputs and instantly classifies flood risk in real time.
Desired Outcome	A Flask web application using XGBoost (96.55% accuracy) that accepts 6 meteorological features and outputs a binary Flood / No Flood prediction with emergency response guidance.
How Might We?	How might we provide disaster management teams with a reliable, real-time ML flood prediction system that enables timely evacuations and saves lives?